

Equation Sheet: Numerical Solution of 2D Incompressible Navier–Stokes

(Dated:)

I. 1. GOVERNING EQUATIONS

Rearranged:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

where $\mathbf{u} = (u, v)$.

$$\left(I - \frac{\Delta t}{2Re} L \right) \mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-\mathcal{N}^{n+1/2} + \frac{1}{2Re} L \mathbf{u}^n \right]. \quad (11)$$

B. Step 2: Pressure Poisson Equation

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (12)$$

II. 2. GRID AND INDEXING

$$\Delta x = \frac{1}{N_x - 1}, \quad \Delta y = \frac{1}{N_y - 1}, \quad (3)$$

$$x_i = i\Delta x, \quad y_j = j\Delta y. \quad (4)$$

Interior nodes: $i = 2, \dots, N_x - 1, j = 2, \dots, N_y - 1$.

C. Step 3: Velocity Correction

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p^{n+1}. \quad (13)$$

III. 3. DISCRETE OPERATORS

Central differences:

$$\frac{\partial u}{\partial x} \approx \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x}, \quad (5)$$

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2}. \quad (6)$$

IV. 4. ADVECTION TERM (AB2)

$$\mathcal{N}^{n+1/2} = \frac{3}{2} \mathcal{N}^n - \frac{1}{2} \mathcal{N}^{n-1}, \quad (7)$$

$$\mathcal{N}_u = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}, \quad (8)$$

$$\mathcal{N}_v = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}. \quad (9)$$

V. 5. FRACTIONAL STEP METHOD

A. Step 1: Intermediate Velocity

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = -\mathcal{N}^{n+1/2} + \frac{1}{2Re} (\nabla^2 \mathbf{u}^n + \nabla^2 \mathbf{u}^*). \quad (10)$$

VI. 6. DISCRETE POISSON EQUATION

$$\begin{aligned} & \frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} \\ &= \frac{1}{\Delta t} \left(\frac{u_{i+1,j}^* - u_{i-1,j}^*}{2\Delta x} + \frac{v_{i,j+1}^* - v_{i,j-1}^*}{2\Delta y} \right). \end{aligned} \quad (14)$$

VII. 7. BOUNDARY CONDITIONS: LID-DRIVEN CAVITY

A. Velocity (Dirichlet)

$$u = 1, \quad v = 0 \quad (\text{top wall}), \quad (15)$$

$$u = 0, \quad v = 0 \quad (\text{other walls}). \quad (16)$$

B. Pressure (Neumann)

$$\frac{\partial p}{\partial n} = 0 \quad \text{on all walls.} \quad (17)$$

Fix gauge:

$$p_{1,1} = 0. \quad (18)$$

VIII. 8. BOUNDARY CONDITION IMPLEMENTATION

A. Velocity: Strong Enforcement

After each step:

- Enforce $\mathbf{u}^*$ at walls
- Enforce $\mathbf{u}^{n+1}$ after correction

B. Ghost Nodes (Velocity)

Example: left wall ($u = 0$)

$$u_{0,j} = -u_{1,j} \quad (19)$$

Top wall ($u = 1$):

$$u_{i,N_y+1} = 2 - u_{i,N_y} \quad (20)$$

General Dirichlet:

$$u_{ghost} = 2u_{wall} - u_{interior} \quad (21)$$

C. Pressure Neumann BC

Example (right wall):

$$\frac{p_{N_x+1,j} - p_{N_x-1,j}}{2\Delta x} = 0 \quad \Rightarrow \quad p_{N_x+1,j} = p_{N_x-1,j} \quad (22)$$

General:

$$p_{ghost} = p_{interior} \quad (23)$$

D. Important: Projection Consistency

Pressure BC must match velocity BC:

$$\frac{\partial p}{\partial n} = \frac{1}{\Delta t}(\mathbf{u}^* \cdot \mathbf{n}) \quad (24)$$

For no-penetration walls:

$$\mathbf{u}^* \cdot \mathbf{n} = 0 \quad \Rightarrow \quad \frac{\partial p}{\partial n} = 0 \quad (25)$$

IX. 9. LINEAR SYSTEMS

$$A_{CN} \mathbf{u}^* = b_u, \quad A_{CN} \mathbf{v}^* = b_v, \quad (26)$$

$$A_p p = b_p. \quad (27)$$

X. 10. CONJUGATE GRADIENT METHOD

$$r_0 = b - Ax_0, \quad (28)$$

$$d_0 = r_0, \quad (29)$$

$$\alpha = \frac{r^T r}{d^T A d}, \quad (30)$$

$$x \leftarrow x + \alpha d, \quad (31)$$

$$r \leftarrow r - \alpha A d, \quad (32)$$

$$\beta = \frac{r^T r}{r_{old}^T r_{old}}, \quad (33)$$

$$d \leftarrow r + \beta d. \quad (34)$$

Stop when:

$$\|r\|_2 \leq \epsilon \|r_0\|_2. \quad (35)$$

XI. 11. STABILITY (GUIDELINE)

$$\Delta t \leq \min \left(\frac{\Delta x}{|u|}, \frac{\Delta y}{|v|}, \frac{Re}{2} \frac{\Delta x^2 \Delta y^2}{\Delta x^2 + \Delta y^2} \right). \quad (36)$$

XII. 12. VALIDATION (GHIA ET AL.)

$$u(x = 0.5, y), \quad v(x, y = 0.5) \quad (37)$$

Compare against benchmark profiles.

XIII. 13. ALGORITHM SUMMARY

1. Compute advection (AB2)
2. Solve Helmholtz problem for $\mathbf{u}^*$
3. Enforce velocity BCs

4. Solve Poisson equation for p
5. Correct velocity
6. Re-enforce velocity BCs
7. Repeat